

A 1280x960 Dynamic Vision Sensor with a 4.95- μm Pixel Pitch and Motion Artifact Minimization

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Abstract—This paper reports a 1280x960 DVS. A 4.95- μm pixel pitch is achieved with in-pixel Cu-Cu connection and the newly designed GIDL-suppression scheme. A sequential column selection scheme and a global event-holding function are implemented to minimize motion artifacts. The power consumption per pixel of 122 nW is 1.25x smaller and the maximum readout speed of 1.3 Geps is 4.33x faster than the previous state-of-the-art.

Keywords—dynamic vision sensor, event-driven, leakage suppression, global event-holding

I. INTRODUCTION

Recently, dynamic vision sensors (DVS) have demanded high resolution for mobile, automotive, and internet-of-things applications, so have tried reducing their pixel pitches to effectively increase the resolution within the limited area. However, the smallest pixel pitch of 9 μm for VGA resolution [1] is still big and hard to increase the resolution and the highest resolution of 1280x800 with the pixel pitch of 9.8 μm [2] is also far from economic mass production. Adopting the more scaled-down technology to shrink pixels brings the severe side effect, especially an increase in leakages. It leads to the degradation of pixel characteristics and a decrease in effective bandwidth of data throughput. Higher resolution causes more motion artifact, introduced by errors in timing accuracy within the limited bandwidth of the event-driven readout. So, the practical implementation of the high-resolution event-driven image sensor poses two serious challenges; 1) how to reduce a pixel pitch and suppress the following side effects and 2) how to eliminate motion artifacts from the timing inaccuracy.

This paper reports a 1280x960 DVS featuring a 4.95 μm . Each pixel circuitry is divided into top and bottom chips with

in-pixel Cu-Cu connection to reduce a pixel pitch. The gate-induced-drain-leakage (GIDL) suppression scheme is newly designed to avoid periodic false event generation. A sequential column selection scheme and a global event-holding function are implemented to minimize motion artifacts.

II. PIXEL DESIGN

A. In-pixel Cu-Cu Connection

A DVS pixel structure is illustrated in Fig. 1. A pixel is composed of a photodiode (PD), a log transimpedance amplifier (LOGA), a source follower (SF), a capacitive-feedback amplifier (CFA), two comparators, their following event storages and in-pixel logic. It results in several tens of transistors for a pixel and its complexity has hindered reducing the pixel pitch to under 9 μm [1] using a more scaled-down process. This constraint was resolved by adopting in-pixel Cu-Cu wafer bonding technology [3] and dividing pixel circuits into top and bottom chips. The circuitry from PD to SF is placed in the top chip, while the rest from CFA are placed in the bottom chip. This allows the pixel pitch to be reduced to 4.95 μm and the pixel area to occupy 24.5 μm^2 . The SF output, a low-impedance node, is chosen for the Cu-Cu interfacing to avoid the leakage current flow between Cu bonding pads. Fig. 2(a) shows the additional Cu bonding pad, which is placed every 2x2 pixels and is reserved for the pixel ground. The minimum Cu-Cu pitch is 4.05 μm between the SF output and the ground nodes. The cross-sectional view of Cu-Cu bonding is also shown in Fig. 2(b).

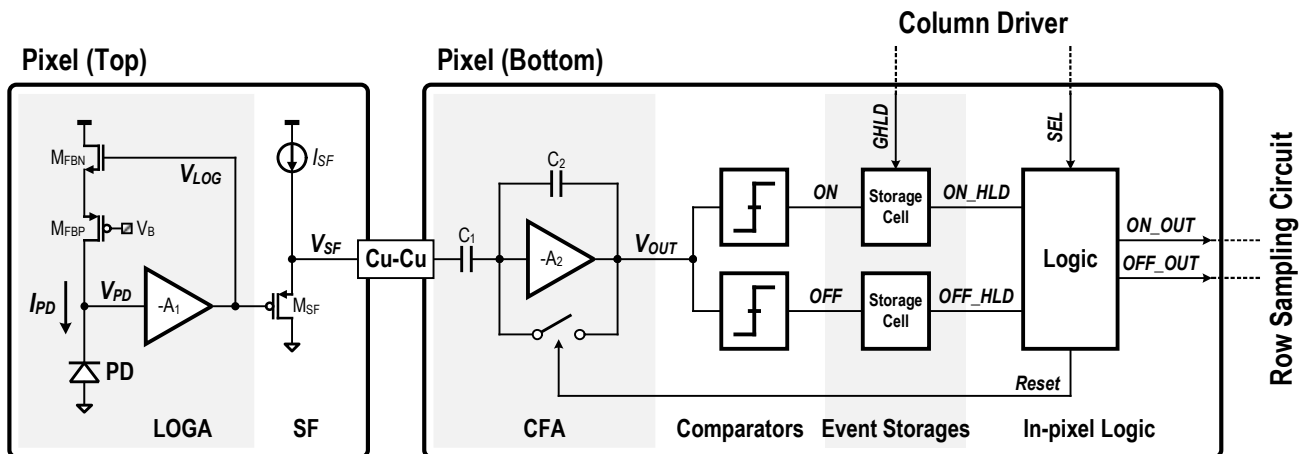


Fig. 1. The proposed pixel structure

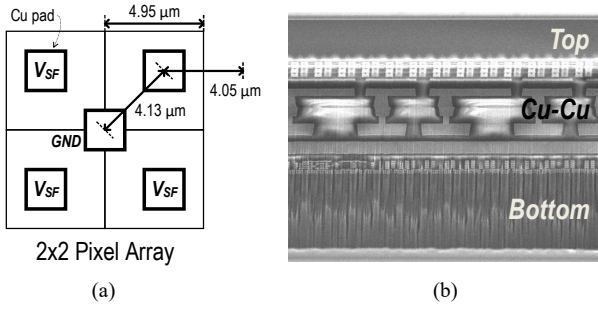


Fig. 2. (a) The placement of 5 Cu bonding pads for 2x2 pixel array, and (b) a cross-section view of Cu-Cu wafer bonding

B. GIDL Suppression

All existing DVSs suffer from the periodic false event generation induced by leakage currents flowing into the floating node [4]. We found out that GIDL is the dominant among the leakage components, while the other leakages are negligible. GIDL is getting worse with more scaled-down technology [5]. Fig. 3 illustrates the conventional reset switch of CFA and how false events are generated periodically. When a pixel operating after reset, the PMOS reset switch is turned off, then V_{IN} node remains floating until the pixel generates an event. The leakage current (I_{GIDL}) charges the floating node (V_{IN}) and CFA output voltage (V_{OUT}) constantly drifts downward. Thus, the pixel periodically generates false ON events without any illuminance changes. The period of false event generation (t_{period}) can be simply derived as (1).

$$t_{period} = V_{EVTH} \cdot (C_2 + C_1/A_2) / I_{GIDL} \quad (1)$$

V_{EVTH} denotes an ON event threshold voltage. For example, 1-fA I_{GIDL} flowing into the floating node generates false events every 400 ms with V_{EVTH} of 100 mV and $(C_2 + C_1/A_2)$ of 5 fF. The voltage drift by GIDL appears as the drift of event thresholds, and distorts the pixel characteristics over time. Therefore, back-end processing for eliminating the periodic false events wastefully consumes bandwidth and decreases efficiency of image processing. The traditional approaches, integrating a high-pass filter into CFA by implementing a high impedance between V_{IN} and V_{OUT} nodes [4], are hard to apply to the GIDL-dominant condition because the implemented impedance is not controllable and vulnerable to PVT variations. The other approach, dealing with the body bias of the reset switch [1], effectively removes the periodic false events but is not suitable because well separation for the body bias control obstructs shrinking pixel area.

Instead of the previously reported schemes, the newly designed GIDL-suppression scheme controls the gate node of the reset switch. Fig. 4 illustrates the structure of the GIDL-suppressed switch and how to suppress GIDL and to remove the switching feedthrough by the proposed scheme. The GIDL-suppressed switch is composed of two reset switches in series and a complementary double source follower. When CFA reset, its reset voltage is sampled at the node between two reset switches (V_{NET}) and during normal operation after reset the source follower biases the sampled reset voltage to the gate node of the reset switches. Thus, the voltage difference ($V_{GS,P}$) becomes almost zero and GIDL current is suppressed to the negligible level, shown in Fig.

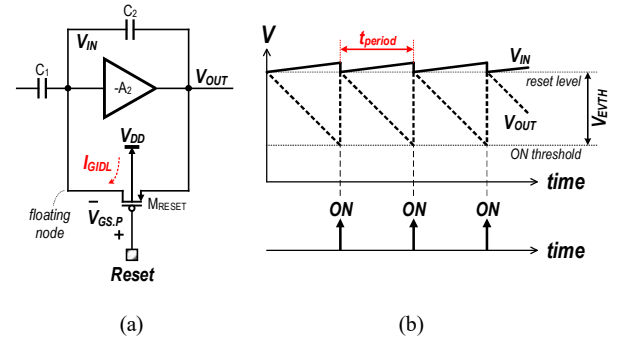


Fig. 3. (a) A conventional PMOS switch of CFA, and (b) the timing diagram of periodic false event generation induced by voltage drift

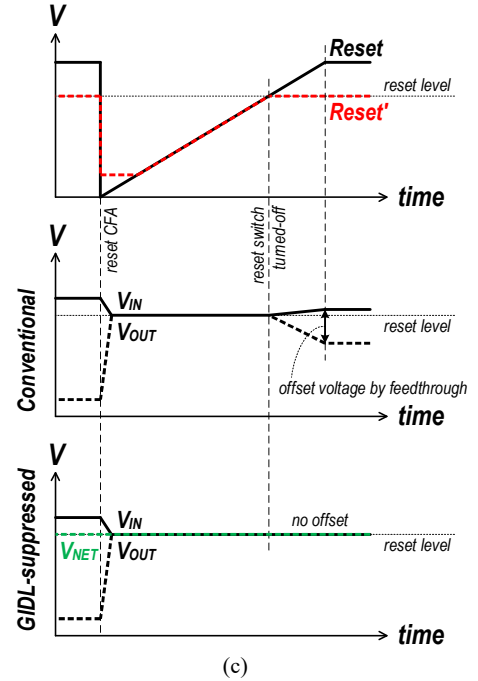
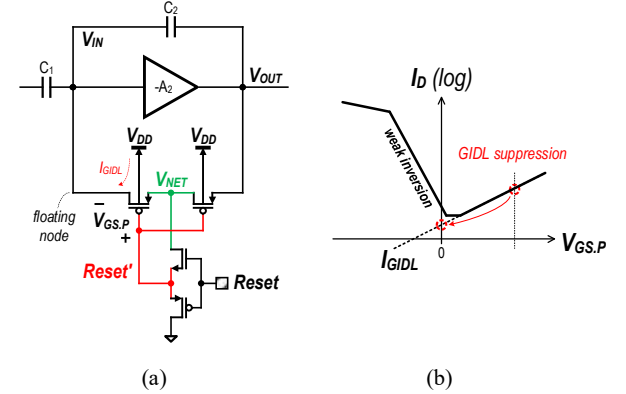


Fig. 4. (a) The GIDL-suppressed reset switch of CFA, (b) I_D - $V_{GS,P}$ curve of M_{RESET} , and (c) the timing diagram of the offset voltage removal by feedthrough when reset switch turning off

4(b). The reset voltage (Reset') is adaptively sampled pixel-by-pixel and time-to-time, so the robustness to PVT variations is successfully achieved.

Furthermore, the zero voltage difference ($V_{GS,P}$), forced by the GIDL-suppressed switch, eliminates the voltage offset on CFA output (V_{OUT}) introduced by the feedthrough at the moment of switching from reset to normal operation, shown in Fig. 4(c).

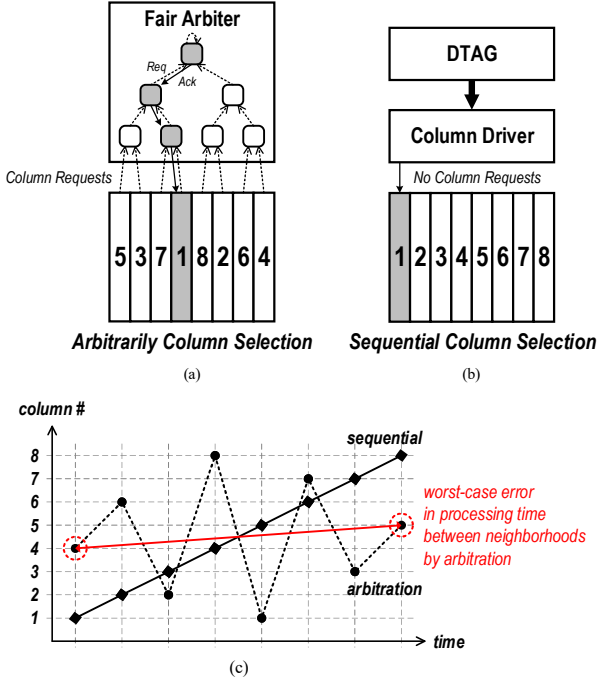


Fig. 5. (a) An example of the arbitrary column selection, (b) an example of the sequential column selection, (c) the comparison of two readout schemes and the timing inaccuracy induced by arbitration

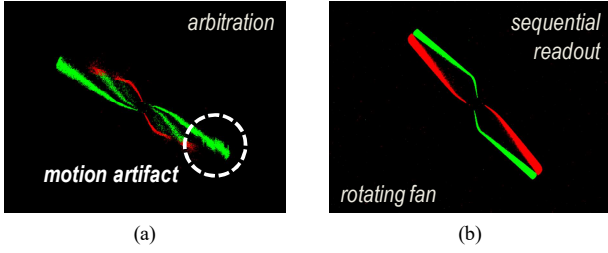


Fig. 6. Two captured scenes of a rotating fan (a) with the arbitrarily column selection, and (b) with the sequential column selection

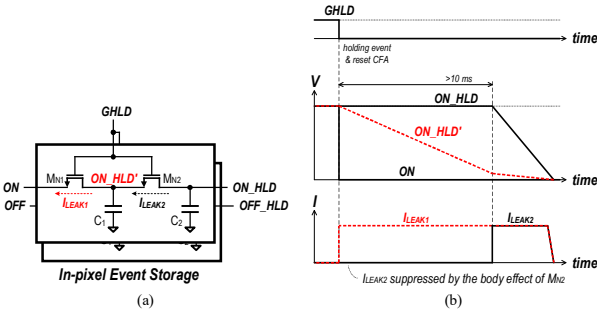


Fig. 7. (a) The structure of integrated storage cells, and (b) the timing diagram of its operation

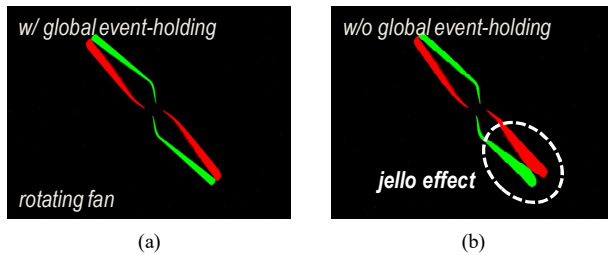


Fig. 8. Two captured scenes of a rotating fan (a) with and (b) without global- event-holding function

A. Sequential Column Readout

A motion artifact is related to its timing accuracy with respect to the subject motion. Fig. 5 illustrates the image artifact introduced by the event-handling order of the fair arbiter [1], compared with the sequential column selection. Each arbiter cell gives an equal chance between two request inputs with first-in-first-out order, and in the same manner, the entire arbiter tree gives equal chances to all of its input requests. As a result, the event-handling order becomes unpredictable and causes the error in processing time even between the neighboring columns which generate events at the same time, shown in Fig. 5(c). Furthermore, the fair arbiter tree adds two undesired latencies between request and acknowledgement; 1) the request and the acknowledgement signals should always travel from bottom to top and from top to bottom of the arbiter tree, respectively, and 2) the metastability in each arbiter cell occurs when two inputs arrive at almost the same time. This motion artifact is solved by replacing the fair arbiter with the column driver which sequentially selects column by column, and thus simply avoids the unpredictable process timing inaccuracy and the undesired latencies in arbitration. Fig. 6 compares two captured scenes by the arbitrarily column selection and the sequential column selection.

B. Global Event-Holding Function

Even with a sequential column selection, when capturing fast motion, another type of the motion artifact known as a jello effect appears because each event for the scene is not captured at exactly the same time. Two in-pixel storage cells are integrated just after comparators and hold an ON or OFF event while scanning the entire array, similar to the global shutter in CMOS image sensor. Fig. 7(a) shows the circuit diagram of the integrated storage cell. Interestingly, the implemented storage cell has a cascaded form of the conventional structure of a switch and a capacitor, for the enough retention time. Fig. 7(b) illustrates how the proposed cascaded structure utilizes a body effect to prevent a channel leakage and holds an event for the required retention

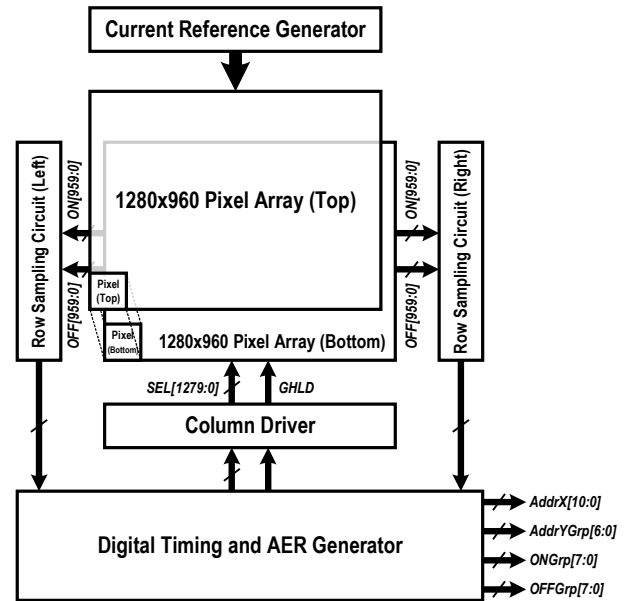


Fig. 9. Top block diagram of the proposed DVS system

TABLE I. COMPARISON WITH THE STATE-OF-THE-ART

	This work	ISSCC 2017 [1]	CVPR 2019 [2]	VLSI 2019 [6]
Process	65 nm 1P6M BSI / 28 nm 1P7M	90 nm 1P5M BSI	65 nm CIS	65 nm 1P9M
Resolution	1280x960	640x480	1280x800	132x104
Chip Area	8.37x7.64 mm ²	8.0x5.8 mm ²	-	2.0x2.0 mm ²
Pixel Pitch	4.95 μ m	9 μ m	9.8 μ m	10 μ m
Fill Factor	22%	-	-	20%
Supply Voltages	2.8 V/1.8V analog 1.0 V digital	2.8 V analog 1.2 V digital	-	1.2 V
Power Consumption ^a (Total)	150 mW ^c	50 mW ^d	400 mW ^c	4.9 mW ^d
Power Consumption ^b (per Pixel)	122 nW	162 nW	391 nW	357 nW
Max Event Rate	1.3 Geps	300 Meps	-	180 Meps
Readout	Sequential	Arbitration	-	Sequential
In-Pixel Storage	Yes	No	-	Yes

a. At max event rate or under high activity condition

b. Normalized by resolution

c. Including power consumption at MIPI as I/O

d. Including power consumption at parallel interface as I/O

time. After an event is sampled on the capacitors, the voltage at ON_HLD' node is continuously drifting toward ground. However, the voltage at ON_HLD node is well secured, because the body effect forces the threshold voltage of M_{N2} increase until the voltage at ON_HLD' node approaches to almost zero. The storage cell is designed to hold an event for >10 ms to support various operation modes. Fig. 8 compares two captured scenes with and without the global event-holding function, respectively.

IV. IMPLEMENTATION AND MEASURED RESULTS

The architecture of the DVS system is shown in Fig. 9. A current reference generator and the front-end part of the pixel array are placed in the top chip, and the back-end part of the pixel array, a column driver, a row sampling circuit and digital circuits are placed in the bottom chip. A row sampling circuit is divided into the left and right sides of the pixel

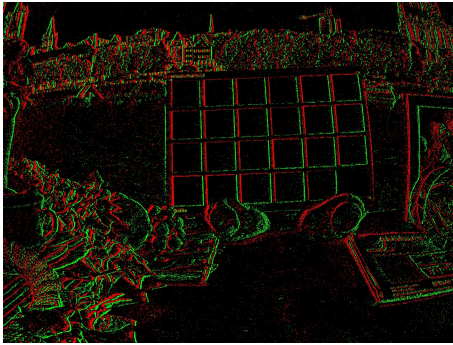


Fig. 10. A complex studio scene

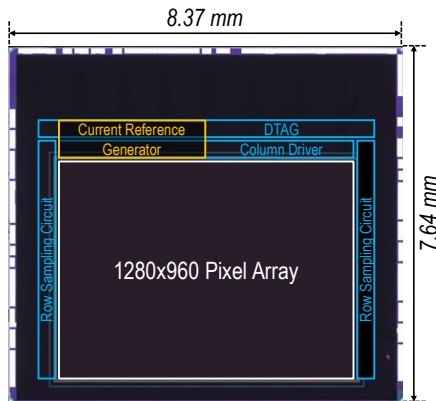


Fig. 11. A chip photo

array to increase readout efficiency. A column driver and a row sampling circuit are simple buffer arrays controlled by the digital timing and address-event representation (AER) generator (DTAG). DTAG generates the sequential column selection signals (SEL) and the global event-holding signal (GHLD) and delivers out only events (ONGrp, OFFGrp) by 8-b grouping and their addresses (AddrX, AddrYGrp) by FastFinder [1].

The measured results are summarized and compared with the state-of-the-art in Table I. The 1280x960 DVS is fabricated in a stack process composed of 65 nm 1P6M BSI process for the top chip and 28 nm 1P7M process for the bottom chip and occupies 8.37x7.64 mm². The power consumption is 150 mW including 2.5-Gbps 4-lane MIPI, and 122 nW per pixel in reduction of 25% from the previous works. The maximum event rate is around 1.3 giga-events per second (Geps), also 4.3x faster than the previous works. Fig. 10 shows a detail of a complex studio scene, captured from the event stream. The fabricated chip photo is shown in Fig. 11.

V. CONCLUSION

A high-resolution DVS of 1280x960 has been implemented. The pixel pitch of 4.95 μ m is achieved with Cu-Cu wafer bonding technology. The GIDL-suppression scheme is newly designed and periodic false event generation is successfully avoided even with the more scaled-down technology. Sequential column selection scheme is employed and global event-holding function is integrated into each pixel to minimize the motion artifacts introduced by the timing inaccuracy.

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